

Rohan Singh

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EDUCATION

University of Virginia Charlottesville, VA
BS in Computer Science – Minor in Data Science May 2027
• Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Computer Systems & Organization

EXPERIENCE

Expedia Group Jun 2026 – Present
Machine Learning Engineer Intern / Python, FastAPI, Trino SQL San Jose, CA
• Built a **FastAPI** prototype for OneKey, a **100M+ member rewards program**, with live profiles and safe fallbacks.
• Proved a **+0.95pp booking lift** in a placebo-clean study of **321k members**, then replicated the dollar-lift result.
• Designed a typed reward API adopted by **2 sibling services**, isolating clients from model changes with safe abstention.
• Defined a **27-field train/serve contract** from 112 model features and verified live intent-data backfill end to end.
• Replay-tested **500k records** across a **100-test suite**, catching eligibility, timeout, and schema failures before demo.

Freddie Mac May 2025 – Jul 2025
Software Engineer Intern / Java, TypeScript, Spring Boot, Angular McLean, VA
• Built **Java Spring Boot** services to validate 120k accounting events and deployed them through **Jenkins CI/CD**.
• Consolidated 200+ REST endpoints behind **GraphQL**, cutting Angular payloads **60%** and simplifying client integration.
• Benchmarked Playwright against Cypress, cutting test time **4x from 8 to 2 minutes** and driving the team’s migration.
• Automated nightly UAT refreshes with Python and Jenkins, cutting load time from **2 hours to 10 minutes per run**.

Biocomplexity Institute Nov 2024 – May 2025
AI/ML Research Assistant / Python, Slurm Charlottesville, VA
• Built Python graph-analysis pipelines with **cuGraph**, SciPy, and NetworkX over 1.2M-edge financial networks.
• Raised AUC-ROC from 0.55 to **0.76**, surfacing **250 anomalies at 89% precision** for downstream investigator review.
• Scaled end-to-end GPU anomaly detection with **cuGraph** to **2M+ nodes and edges** on UVA’s Slurm cluster.

Cubic Corporation Jun 2024 – Aug 2024
Software Engineer Intern / JavaScript, React, Terraform Sterling, VA
• Built and deployed an AWS-hosted React/Docusaurus developer portal with Terraform, serving **500+ engineers**.
• Cut onboarding tickets **85%, from 90 to 13 in one month**, by redesigning site flow and navigation after user tests.
• Added Algolia DocSearch and simplified navigation, reaching **95% top-3 accuracy at 120ms median query latency**.
• Integrated AWS CodeBuild lint and test checks on pull requests, reducing manual QA and speeding team merges.

PROJECTS

Shortlist | Go, Python, PyTorch, Next.js Dec 2025 – Jul 2026
• Served exact retrieval across **87,585 movies** in Go with float16 embeddings and history filtering at **4.0ms median**.
• Raised **HitRate@10 from 73.8% to 84.1%** by aligning training with the product’s 1–5 favorite onboarding flow.
• Shipped a Next.js app on Vercel with multi-title onboarding, natural reasons, saved lists, and **Playwright/Axe CI**.

Marathon Prep Bot | TypeScript, Node.js, Garmin FIT SDK Jun 2026 – Present
• Built a strict TypeScript pipeline that decodes Garmin FIT binaries into **1,300+ CRC-validated activities**.
• Automated 6-hour iCloud syncs with **202 regression tests across 21 suites**, keeping GPS and health data local.
• Shipped a live dashboard turning 13 months of activity data into weekly plans, training-load trends, and daily guidance.

Parking Shark | Node.js, Express, MySQL, GCP Jan 2026 – Apr 2026
• Built a Node/Express parking marketplace on GCP with listing, reservation, payment, and review flows in a 4-person team.
• Designed a **13-table MySQL** schema with locking, overlap triggers, and atomic transactions that prevent double-booking.
• Added MySQL-backed CI and integration tests for authentication, ownership, and state transitions across every route.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Go, C++, SQL
Frameworks: Spring Boot, Node.js, Express, React, Next.js, FastAPI, GraphQL, PyTorch
Infrastructure & Data: AWS, GCP, Docker, Terraform, Jenkins, GitHub Actions, PostgreSQL, MySQL, MongoDB